Applying the Critical Velocity Model for an Off-Season Interval Training Program

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Abstract

Clark, IE, West, BM, Reynolds, SK, Murray, SR, and Pettitt, RW. Applying the critical velocity model for an off-season interval training program. *J Strength Cond Res* 27(12): 3335–3341, 2013—The critical velocity (CV) model offers an opportunity to prescribe and to test empirically different velocity-time (V–t) configurations of high-intensity interval training (HIIT); however, such experiments are lacking. We evaluated a group of competitive, female soccer players (age = 19 ± 1 years, height = 168 ± 6 cm, mass = 61 ± 6 kg) completing 1 of 2 different HIIT regimes: a short group ($n = 6$) completing higher V and shorter t configurations, and a long group ($n = 10$) completing lower V, longer t configurations. Both groups trained 2 d·wk^{-1} for 4 weeks. For each workout, both groups ran at velocities exceeding CV and designed to deplete identical fractional percentages of the finite work capacity above CV (D'). The metrics of CV and D' were evaluated at pretraining and posttraining using the 3-minute all-out exercise test on an indoor track using video digitizing of displacement relative to time. Despite differences in the V–t configurations, both groups increased their CV (+0.22 m·s^{-1}, +6%) and decreased their D' (−24 m, −13%; $p < 0.05$). We conclude that 2- to 5-minute HIIT bouts are suitable for increasing CV, in previously trained athletes, but they result in a decline of D'. To increase D', we suggest examining HIIT of intensities that are <2 minutes and >130% of maximum oxygen uptake.

Key Words 3-minute all-out exercise test, anaerobic power, critical power, high-intensity interval training

Introduction

Conditioning exercise is typically rhythmic and repetitive (e.g., running and cycling) and viewed primarily as a mode of exercise for enhancing aerobic fitness. Conditioning exercise, if carried out at appropriate intensities, can enhance an athlete's capacity for high levels of sustained work (7,16,19). One common conditioning model for improving anaerobic capacity and maximum oxygen uptake ($\dot{V}O_2max$) is high-intensity, interval training (HIIT) (1).

Interval training is often prescribed as a percentage of the power or velocity evoking maximum oxygen uptake ($\dot{V}O_2max$) or maximum heart rate (HRmax) (1). Such prescriptions do not take into consideration the between-subject differences in anaerobic capacity. The critical velocity (CV) model (6) offers a unique solution to the problem. First, the parameter CV identifies the lower limit of exercise necessary for evoking $\dot{V}O_2max$ (15). Specifically, when an athlete runs at a constant velocity exceeding CV, $\dot{V}O_2$ will continue to rise toward and achieve maximum, if given sufficient time (8). Second, the CV model also identifies D' (pronounced D prime). The D' is the displacement capacity of the body at velocities greater than CV (6). For example, if an athlete with a CV of 4 m·s^{-1} and a D' of 180 m were to run at 5 m·s^{-1}, his or her time limit to exhaustion (t_{LIM}) at that velocity would be 180 seconds, or the product of 1 m·s^{-1} and 180 m.

The quantification of CV and D' traditionally required determining t_{LIM} (x-axis) for a series of different running distances (y-axis) (6). Linear regression can identify an unknown distance for a given t_{LIM}, where the slope is CV, and the y-intercept is D' (12). The 3-minute all-out exercise test (3 MT) (3,18) was validated recently for running (13), and it eliminates the need for multiple time trials of different distances. Specifically, the running 3 MT presumes that the displacement and t_{LIM} at 150 and 180 seconds of all-out running can reveal CV and D'. The average velocity in the 3 MT between 150 and 180 seconds is presumed to equal CV, because 150 seconds is deemed a sufficient duration to deplete D' wholly. The D' also can be derived as 150 seconds (V_{150s} − CV), where V_{150s} is the velocity for the initial 150 seconds of all-out running.

The application of the CV model to HIIT offers the advantage of prescribing intervals based on a percentage of D', relative to each athlete's CV (9,11). Target times for given distances (track intervals) or times for given velocities (treadmill intervals) can be determined by modifying 2-component model equations (6,20).

Despite the capacity to model identical expenditures of D' for different distances and time, shorter intervals at the higher running velocity (V) hypothetically would serve to enhance training the anaerobic capacity. Conversely, longer

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intervals at the lower running V hypothetically would serve to enhance training the CV. Unfortunately, the precise delineation of V and interval time (Int_t) configurations to selectively train and improve D' or CV are unavailable. Therefore, the purpose of this study was to evaluate the effects of 2 separate HIIT regiments aimed at enhancing CV or D', respectively.

Methods

Experimental Approach to the Problem

A group of female soccer players completed an off-season interval training program designed using the CV model. The parameters of CV and D' were determined using a running 3 MT (13) and used to model velocities corresponding to given distances (equation 1). The players were ranked at pretesting and assigned either to a group aimed at increasing CV or D', dependent upon their training needs relative to their ranking on the team. The post 3 MT was conducted subsequent to a 4-week training period to determine the difference between groups for changes in CV and D'. All testing and HIIT sessions were conducted in the morning hours. No modifications in the team's hydration, dietary, and sleeping habits were made for the purposes of this study.

Subjects

An original sample of 20 subjects from a National Collegiate Athletic Association Division II Women's Soccer program was tested preliminarily and divided into either a short or long HIIT group based on their 3 MT. All the subjects were members on the team previously and were intending to return for the upcoming season. From the original sample, a total of 16 women completed all training and testing trials: 10 women were assigned to a long group, and 6 women were assigned to a short group, based on the pretesting performance. We refer to this subsample of 16 subjects as the "team" in our results. The demographic data for the team are as follows: age = 19 ± 1 years, height = 168 ± 6 cm, mass = 61 ± 6 kg (N.B., no significant between-group differences, $p < 0.05$). Each subject was healthy and free of injury, as determined from a preliminary physician's examination, and each declared her abstinence from performance-enhancing drugs as part of her athletic eligibility. The subjects were informed of the purpose of the study and provided informed consent to participate. The procedures of this study were previously approved from our university's institutional review board for research.

All the subjects took part in a supplemental resistance exercise program for 3 $d \cdot wk^{-1}$. These training sessions consisted primarily of multiple joint explosive and heavy resistance lifts, with each of these sessions not occurring on days that HIIT sessions were scheduled. The load assigned for each lift ranged 60–85% of their individual 1-repetition maximum.

Three-Minute All-Out Exercise Test Procedures and Outcome Data

The subjects completed a pretesting and posttesting running 3 MT (13). The original version of this 3 MT involved the

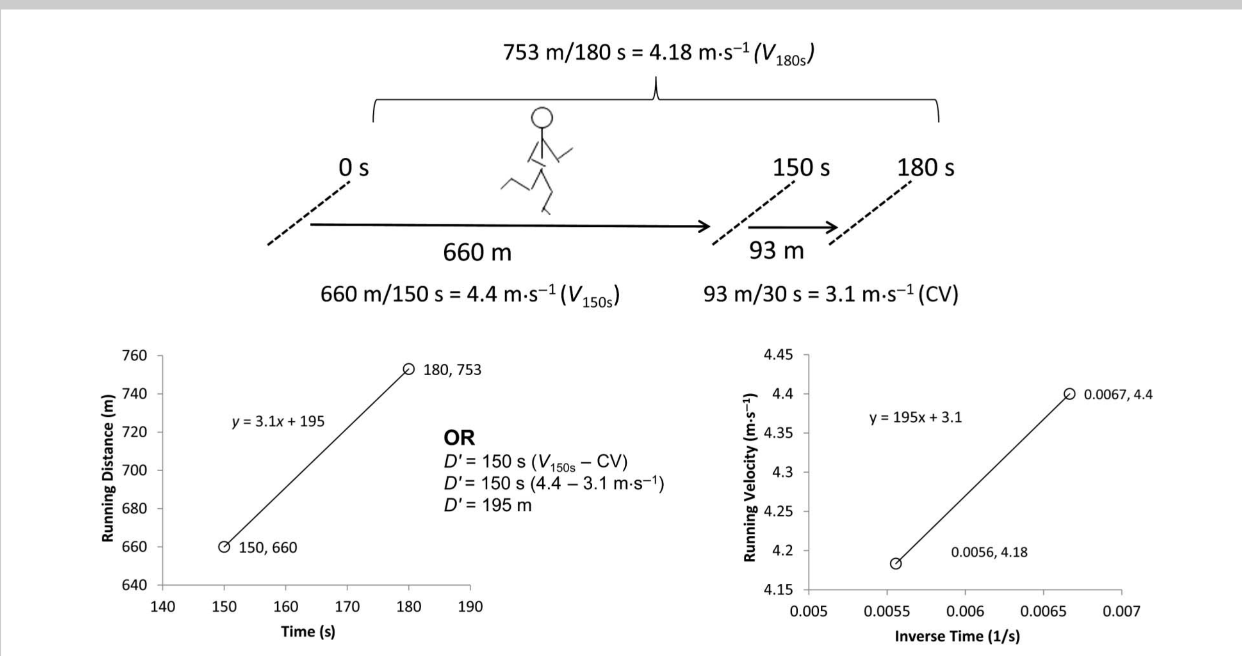

Figure 1. Modeling of critical velocity (CV, 3.1 $m \cdot s^{-1}$) and D' (195 m) for a representative subject using linear regression analysis of data from the 3-minute all-out exercise test. Note that her CV also is resolved from the velocity between 150 and 180 seconds, and that D' also is resolved using 150 seconds (V_{150s} − CV).

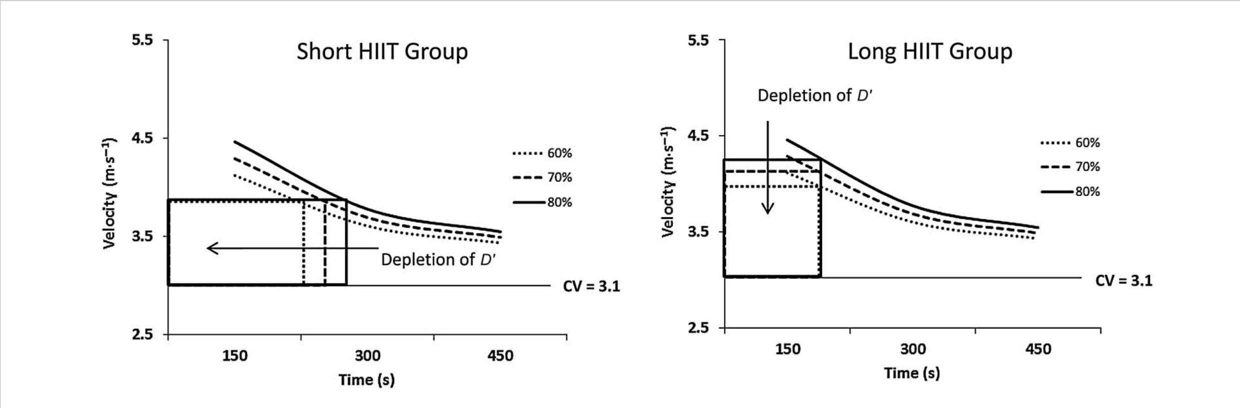

Figure 2. Modeling of velocity-time curves for the subject in Figure 1 illustrating high-intensity interval training (HIIT) bouts using 2 different methods of depleting 3 different percentages of D'. The left panel depicts the method of maintaining velocity and reducing time and distance (i.e., what we defined as our short HIIT group). The right panel depicts the method of reducing velocity for a given distance (i.e., what we defined as our long HIIT group).

use of global positioning sensor (GPS) data; however, we conducted our testing on an indoor, 200-m field-house track. We recorded performances by digital video at a rate of 30 Hz (Flip Video, Pure Digital Technologies, USA; N.B., sampling rates for GPS are much slower with sampling at every 3 seconds). Starting and ending times on the track were monitored by a stopwatch for 3 minutes and additional 5 seconds to ensure full video data collection. Orange cones were placed every 10 m as way points, and the subjects ran on the inside lane. The camera was tripod mounted from an overhead view, but some panning was necessary to keep the subject in view throughout the test. Commercial video editing shareware (Windows Live Movie Maker, Microsoft Corporation, Remond, WA, USA) was used to retrieve the times at every 10 m. The displacements at 150 and 180 seconds were interpolated from the surrounding way points. For example, if the elapsed time at the 710-m waypoint was 149 seconds and at 720 m was 151 seconds, displacement at 150 seconds was interpolated as 715 m. The procedure for evaluating CV and D' is illustrated in Figure 1.

The normal pacing strategy for a time trial is to start out with an extreme velocity (i.e., far beyond what can be sustained by aerobic metabolism), decelerate to constant velocity in close proximity to CV, and then end with a spurt of all-out acceleration (4). We instructed our subjects to run all out for the entire test and to avoid pacing. Encouragement was provided, but the subjects were neither informed of elapsed time nor time remaining. We also evaluated the individual velocity-time (V–t) records to ensure the appearance of a velocity nadir as opposed to a sudden, spurt of acceleration at the end of the 3 MT, indicative of pacing.

In prior HIIT studies (7,16,19) where critical power and the power–t_{LIM} curvature constant (W') were measured at pretesting and posttesting, the interval intensities were prescribed at a percentage of $\dot{V}O_2$max (~100–105%). The investigators of the running 3 MT determined that 90 seconds of all-out exercise approximated the velocity of $\dot{V}O_2$max ($v\dot{V}O_2$max) (13). For comparative purposes, we estimated the $v\dot{V}O_2$max from the velocity at 90 seconds to allow us to determine the percentages of the $v\dot{V}O_2$max at which our subjects ran their intervals.

Interval Prescriptions

Using a prescription depleting 80% of D' as an example interval, the equations for each scenario are expressed as follows:

$$V_t = [(D' \times 0.8)/t_{LIM}] + CV, \tag{1}$$

$$Int_t = [D - (D' \times 0.8)]/CV, \tag{2}$$

$$Int_t = (D' \times 0.8)/(V - CV), \tag{3}$$

where V_t is the interval velocity, Int_t is the interval time, D is the distance, and V is the velocity. An example prescription for depleting different percentages of D' for different Int_t values are shown for the short and long HIIT bouts in Figure 2 (N.B., for this example, we used the data for the subject appearing in Figure 1). For instance, her peak velocity and time for running 1,000 m would be modeled at 4.16 $m \cdot s^{-1}$ and 241 seconds, respectively. If she ran 800 m within 193 seconds at that same velocity of 4.16 $m \cdot s^{-1}$ (equations 2 or 3), she would deplete 80% of D' (Figure 2, left panel). Or, the velocity at 80% of the difference between 4.16 and 3.1 $m \cdot s^{-1}$ (i.e., CV) would be set for a 1,000-m interval depleting 80% of D' (Figure 2, right panel).

High-Intensity Interval Training Exercise Protocols

After attaining the subjects' CV and D', we calculated each subject's specific Int_t. The subjects were assigned to either a short or long group depending on their lack of CV or D'

TABLE 1. High-intensity interval training programs for the short and long groups.*

	Week 1		Week 2		Week 3		Week 4	
Intensity	Thursday	Saturday	Thursday	Saturday	Thursday	Saturday	Thursday	Saturday
Interval percent	60	60	70	60	80	70	80	Practice scrimmage
Short group	4 × 600 m	4 × 600 m	3 × 800 m	4 × 800 m	3 × 600 m	3 × 600 m	3 × 800 m	
%$v\dot{V}O_2max$	118	118	115	112	127	122	118	
Long group	4 × 800	4 × 800	3 × 1,000	4 × 1,000	3 × 800	3 × 800	3 × 1,000	
%$v\dot{V}O_2max$	102	102	101	98	110	106	104	

*Prescriptions represent a percentage of D' derived from equations 1–3 (see Figure 2 for a visual comparison). Percentages of the estimated $v\dot{V}O_2max$ are estimates based on the V_{90s} measurement from the 3-minute all-out exercise test.

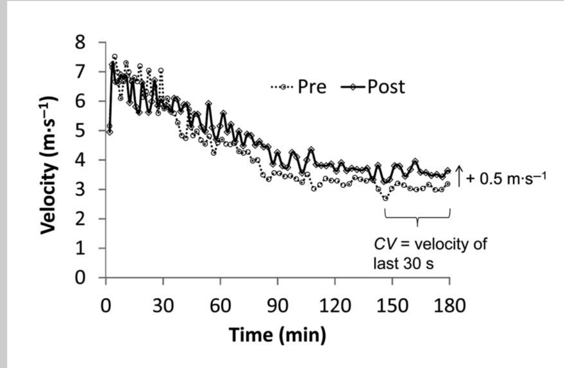

Figure 3. Representative 3-minute all-out exercise test at pretesting and posttesting for representative subject (note same subject in Figures 1 and 2).

relative to the team's mean CV and D'. Specifically, the short group had lower D' values relative to the team average, whereas the long group had lower CV values relative to the team average. Our aim with this 4-week training protocol was to make the team more homogeneous for both CV and D' parameters before implementing a new phase of training. We chose a 4-week period because this time frame has been reported to evoke improvements in critical power for cycle ergometry (19).

The team completed HIIT intervals for 2 $d \cdot wk^{-1}$ for a 4-week period (Table 1). We point out that Table 1 was not an a priori program; it evolved based on our observations of the team's response to each workout. Each subject wore chest straps to measure the HR, and the data from each workout were downloaded (Polar Protrainer 5, Polar Products, Lake Success, NY, USA). To avoid disrupting the flow of the workout by questioning each athlete after each

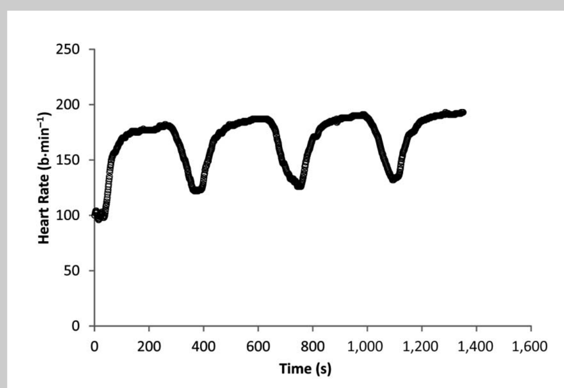

Figure 4. Heart rate (HR) responses for a representative subject to a given high-intensity interval training prescription. Take note of the absence of the steady state for each interval and that end-exercise HR increased progressively despite no change in velocity-time demands.

Table 2. Pretesting and posttesting descriptive statistics (mean ± *SD*) from the 3-minute all-out exercise test.*†

Parameter	Short HIIT group ($n = 6$) Pre	Post	Post-Pre *d*	Long HIIT group ($n = 10$) Pre	Post	Post-Pre *d*	Total team ($N = 16$) Pre	Post	Post-Pre *d*
$v\dot{V}O_2max$ ($m \cdot s^{-1}$)	3.65 ± 0.25‡	3.82 ± 0.21‡	+0.75§	3.71 ± 0.25‡	3.82 ± 0.21‡	+0.51§	3.68 ± 0.24‡	3.82 ± 0.21‡	+0.62§
CV ($m \cdot s^{-1}$)	3.53 ± 0.24‡	3.78 ± 0.25‡	+1.02§	3.11 ± 0.20‡	3.32 ± 0.23‡	+0.97§	3.27 ± 0.29‡	3.49 ± 0.32‡	+0.72§
D' (m)	176.5 ± 15.2‡	146.8 ± 40.9‡	−1.06§	234.9 ± 34.7‡	213.9 ± 32.6‡	−0.62§	213.0 ± 40.6‡	188.8 ± 48.2‡	−0.55§
V_{180s} ($m \cdot s^{-1}$)	4.28 ± 0.20‡	4.38 ± 0.22‡	+0.49§	4.15 ± 0.21‡	4.27 ± 0.20‡	+0.60§	4.20 ± 0.21‡	4.31 ± 0.21‡	+0.55§

*HIIT = high-intensity interval training; CV = critical velocity.
†Effect size differences of posttesting minus pretesting (Post-Pre *d*) reflect the mean difference divided by the pooled *SD*.
‡Significant ($p < 0.05$) group differences. There were no significant interactions.
§Significant ($p < 0.05$) pretesting to posttesting difference.

interval, we randomly asked some subjects upon completion of the interval to determine their rate of perceived exertion (RPE) (2). We used Thursdays to introduce new interval workouts to the subjects. Each subject was provided a specific Int_t along with a target lap time to assist with pacing. A large digital clock was in view, and encouragement was provided. On Saturdays, the subjects were given the option to run either track intervals or treadmill intervals, based on the request of the coaches (i.e., some subjects preferred completing intervals on the treadmill). Treadmill workouts were calculated to the specific velocity for a given time period (equation 1). The conversion of treadmill speed and grade was determined from the table appearing in Pettitt et al. (13).

Statistical Analyses

Separate group (short vs. long) by time (pretesting vs. posttesting) factorial analyses of variance were conducted on measures of CV, D', $v\dot{V}O_2max$, and V_{180s}. Effect size (ES) from pretesting to posttesting was calculated using Cohen's *d* (mean difference/pooled *SD*). Rank ordering on the measures from the 3 MT from pretesting and posttesting was determined using Pearson's product-moment correlation coefficients (*r*). Summary statistics are reported as mean ± *SD* ($M \pm SD$).

Results

Visual inspections of the V–t records from the 3 MTs were similar to those gathered using GPS technology. The achievement of a nadir in V at the end of the 3 MTs indicated that the subjects did not pace. A representative subject's results at pretesting and posttesting appear in Figure 3. Fifteen of the 16 subjects were able to achieve or surpass their Int_t on a systematic basis (N.B., the lone subject not able to achieve her Int_t was in the long HIIT group, so her exclusion would not affect our statistical results). The

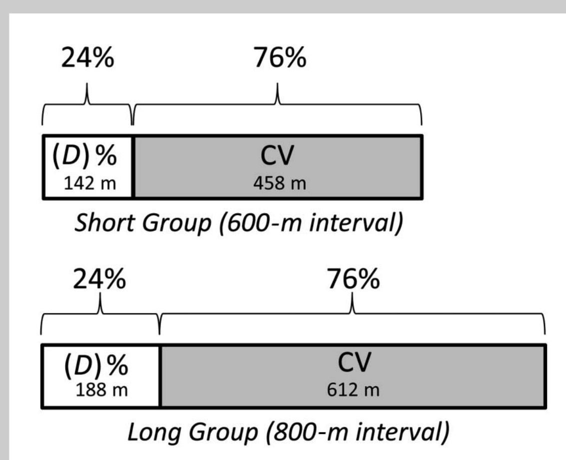

Figure 5. Modeling of intervals using a proportion of D' relative to total interval distances for the mean data of the short ($n = 6$) and long ($n = 10$) groups.

subjects evaluated statistically ($n = 16$) completed every workout along with the pretesting and posttesting 3 MT. Exercising HR data for the HIIT workouts corroborated that the intervals evoked progressively higher end-exercise HR values with each successive interval (Figure 4). The subjects also reported higher RPEs with each interval.

No significant interactions were observed between groups from pretesting to posttesting for CV ($F = 0.15$, $p = 0.70$) and for D' ($F = 0.18$, $p = 0.68$), respectively. Both groups remained significantly different from each other at pretesting and posttesting for CV ($F = 15.6$, $p < 0.01$) and for D' ($F = 6.21$, $p = 0.03$), respectively. The team as a whole exhibited an increase of CV by 0.22 $m \cdot s^{-1}$ ($F = 4.10$, $p < 0.01$) and a decrease in D' by 24 m ($F = 2.53$, $p = 0.02$, Table 2). The $v\dot{V}O_2max$ for the team increased from pretraining to posttraining by 0.14 $m \cdot s^{-1}$ ($F = 2.49$, $p = 0.03$), and V_{180s} increased by 0.11$m \cdot s^{-1}$ ($F = 3.98$, $p < 0.01$).

Strong positive correlation coefficients for pretesting and posttesting measures were observed for CV ($r = 0.76$, $p < 0.01$) and V_{180s} ($r = 0.85$, $p < 0.01$), respectively. The pretesting and posttesting measures for $v\dot{V}O_2max$ ($r = 0.51$, $p = 0.04$) and D' ($r = 0.64$, $p = 0.02$), respectively were positive and moderate.

Discussion

The principal findings of our study are as follows. Four weeks of HIIT were sufficient to evoke moderate, ES improvements in CV in trained soccer players. An unexpected finding was that both groups experienced reductions in D'. Thus, shorter and more intense HIIT bouts may be necessary to evoke improvements in D'. We believe the results of our study are a first step in solving the issue of how to evoke improvements in D' while maintaining CV.

When using the CV model to prescribe HIIT, each interval is projected to evoke progressively higher metabolic responses (i.e., progressively higher $\dot{V}O_2$ values), despite the assignment of an identical magnitude of work (1). The relationship between HR and $\dot{V}O_2$ responses in exercise bouts exceeding gas exchange threshold is fairly equivalent (i.e., the greater the magnitude greater than CV, the more pronounced the time-dependent rise in HR and $\dot{V}O_2$ response) (14). In addition to maintaining targeted times, we verified that the subjects' end interval HR responses were progressively higher with each consecutive bout. The subjects also indicated that the intervals were progressively more difficult, with each successive bout, based on RPE.

The net result of improved CV (Table 2) would indicate that the protocols outlined in Table 1 were valid for the goal of improving CV in trained athletes. The factors contributing to CV, or critical power in cycling, include those factors mediating the maximal lactate steady state (MLSS) and $\dot{V}O_2$ steady state (15). Specifically, intensities exceeding the CV result in continually rising blood lactate values greater than the MLSS along with a time-dependent rise in $\dot{V}O_2$ toward $\dot{V}O_2max$ (3,15). Prior research indicates that HIIT will increase $\dot{V}O_2max$ and $v\dot{V}O_2max$ (7,16,17,19), and our findings of improved CV and estimated $v\dot{V}O_2max$ support the concept of using HIIT to improve aerobic power.

The HIIT durations we selected for this study were based upon the recommendations of Daniels (5). He recommended a range of HIIT between 2 and 5 minutes, where intervals >5 minutes were deemed too slow, and 2-minute intervals were deemed an insufficient duration to allow stressing the "aerobic system." Our concern at the onset of the study with our short group (Table 1) was assigned HIIT bouts too short (~2 minutes) to maintain CV. In retrospect, it appears approximately 2 minutes up to 5 minutes for HIIT has a comparable effect on CV. Moreover, it would appear that HIIT bouts >2 minutes are of too low an intensity to evoke improvements in D'.

A very plausible explanation for our unexpected response of D', unforeseen at the onset of the study, was the influence of the percentage contribution of D' to an interval bout may influence the trainability of D'. For example, our short and long groups had mean D' values of 177 and 235 m, respectively. Therefore, 80% of those D' values would be 142 and 188 m. If the short group ran 600-m intervals and the long group ran 800 m for 80% intervals, the D' contribution to intervals was equivalent at 24% (i.e., 142 m/600 m and 188 m/800 m, respectively). The aerobic system contribution (i.e., 76%) to the intervals also was identical for each group despite running at 2 different velocities and distances (Figure 5), explaining potentially why we observed similar improvements in CV between groups. Future investigators using the CV model to prescribe HIIT programs may wish to intentionally stagger differences in the percentage of D' to the total energy being performed in the interval. A hypothesis worth testing is that a higher percentage of D' to a total interval is necessary to evoke a training effect on D'. In other words, for D' to improve, D' must contribute more to the total distance by running shorter distances and at faster velocities.

Our findings of improved CV with a decline in D' are similar to previous cycle ergometry HIIT studies demonstrating improved critical power with a decline in W' (7,16). These earlier studies used lower HIIT intensities of 100–105% $\dot{V}O_2max$, with 10 repetitions and 2-minute durations. A more recent HIIT study by Vanhatalo et al. (19) used six 5-minute intervals per week at 105% of estimated critical power along with 1 weekly session of 10 × 2 minutes, where the intervals were prescribed to deplete 50% of W'. These investigators also observed improvements in critical power with no differences in W'. As indicated in Table 1, our short group ran intervals between 118 and 127% $v\dot{V}O_2max$, intensities >100–105% $\dot{V}O_2max$. Thus, more extreme intensities (e.g., >130% $\dot{V}O_2max$) are likely necessary to evoke improvements in D'.

We observed improvements in $v\dot{V}O_2max$ from HIIT, as estimated from V_{90s} of the 3 MT. The estimate would seem reasonable as a method of determining a customized graded exercise test duration, as originally suggested (13); however, our correlation analysis indicated subjects shifted rank-order considerably on $v\dot{V}O_2max$. Thus, the V_{90s} estimate method for estimating $v\dot{V}O_2max$ is a poor metric for evaluating training-induced responses, in comparison to CV or total performance, as measured by V_{180s}. The disparity of V_{90s} may relate to the variability of a subject's anaerobic capacity. Prior research indicates when glycogen stores are depleted, anaerobic capacity is markedly reduced (10). Our lack of control for dietary intake is acknowledged as a limitation in this regard.

Practical Applications

With the advancement of the running 3 MT (13), determining CV and D' is relatively simple. Although the critical power or the CV model often is applied for predicting maximal performances, a greater utility for the model is to prescribe HIIT. This study, to our knowledge, is the first study to apply the CV model to prescribe and evaluate a running HIIT program. We recommend the use of the critical power or CV model for prescribing HIIT. In prior trained athletes, we demonstrated that intervals of 60–80% D' depletion were sufficient to evoke improvements in overall running performance (i.e., V_{180s}) along with aerobic power, as measured by an estimate of CV. Shorter durations and higher intensities (i.e., <2 minutes and >130% $\dot{V}O_2max$), may be necessary for evoking improvements in D'.

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